

TuniCars+

Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

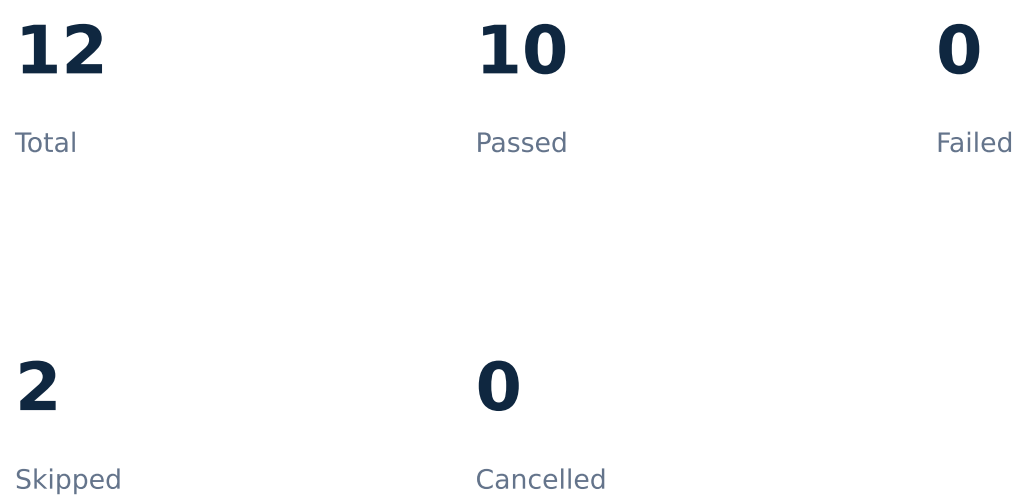
20260813_012315_7c5095

Date

2026-08-13T00:23:15.166624+00:00

Executive Summary

20260813_012315_7c5095



Overall measured availability: 44.02%

During this campaign, 10 of 12 scenarios passed, 0 failed and 2 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	kubernetes	swarm
container-kill	1	1
cpu	1	1
latency	1	1
memory	1	1
network-partition	1	1
node-failure	1	1

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
container-kill	swarm	3.407	17.969	True	True	12.907	64.435	16	11.11	PASS	N/A	N/A
node-failure	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot demonstrate real node	N/A
cpu	swarm	N/A	N/A	N/A	Not Required	64.377	481.197	N/A	100.0	PASS	N/A	N/A
memory	swarm	N/A	N/A	N/A	Not Required	12.783	20.717	N/A	100.0	PASS	N/A	N/A
network-partition	swarm	0.11	0.047	True	True	27.115	67.7	15	N/A	PASS	N/A	N/A
latency	swarm	4.563	5.015	True	True	4518.298	4534.676	11	N/A	PASS	N/A	N/A
container-kill	kubernetes	10.156	20.484	True	True	1244.053	2048.147	7	30.0	PASS	N/A	N/A
node-failure	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	kubelet inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL	N/A
cpu	kubernetes	N/A	N/A	N/A	Not Required	276.538	889.874	N/A	100.0	PASS	N/A	N/A
memory	kubernetes	94.078	3.344	True	True	191.438	636.955	1	99.11	PASS	N/A	N/A
network-partition	kubernetes	0.437	0.328	True	True	163.104	999.306	14	N/A	PASS	N/A	N/A
latency	kubernetes	4.671	5.171	True	True	4593.653	4745.217	11	N/A	PASS	N/A	N/A

swarm — container-kill

State: Pass
Detection Time (s): 3.407
Recovery Time (s): 17.969

HTTP Success: N/A

HTTP Failures: 16

Availability (%): 11.11

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — node-failure

State: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Single-node Docker Swarm cannot demonstrate real node rescheduling.

swarm — cpu

State: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — memory

State: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — network-partition

State: PASS
Detection Time (s): 0.11

Recovery Time (s): 0.047

HTTP Success: N/A

HTTP Failures: 15

Availability (%): N/A

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — latency

State: Pass
Detection Time (s): 4.563

Recovery Time (s): 5.015

HTTP Success: N/A

HTTP Failures: 11

Availability (%): N/A

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — container-kill

State: Pass
Detection Time (s): 10.156

~~Recovery Time (s): 20.484~~

HTTP Success: N/A

HTTP Failures: 7

Availability (%): 30.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — node-failure

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

kubernetes — cpu

Status: Pass
Detection Time (s): N/A
Recovery Time (s): N/A
HTTP Success: N/A
HTTP Failures: N/A
Availability (%): 100.0
CPU (%): N/A
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — memory

Status: Pass
Detection Time (s): 94.078

Recovery Time (s): 3.344

HTTP Success: N/A

HTTP Failures: 1

Availability (%): 99.11

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — network-partition

Status: Pass
Detection Time (s): 0.437

Recovery Time (s): 0.328

HTTP Success: N/A

HTTP Failures: 14

Availability (%): N/A

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — latency

State: **Pass**
Detection Time (s): 4.671

Recovery Time (s): 5.171

HTTP Success: N/A

HTTP Failures: 11

Availability (%): N/A

CPU (%): N/A

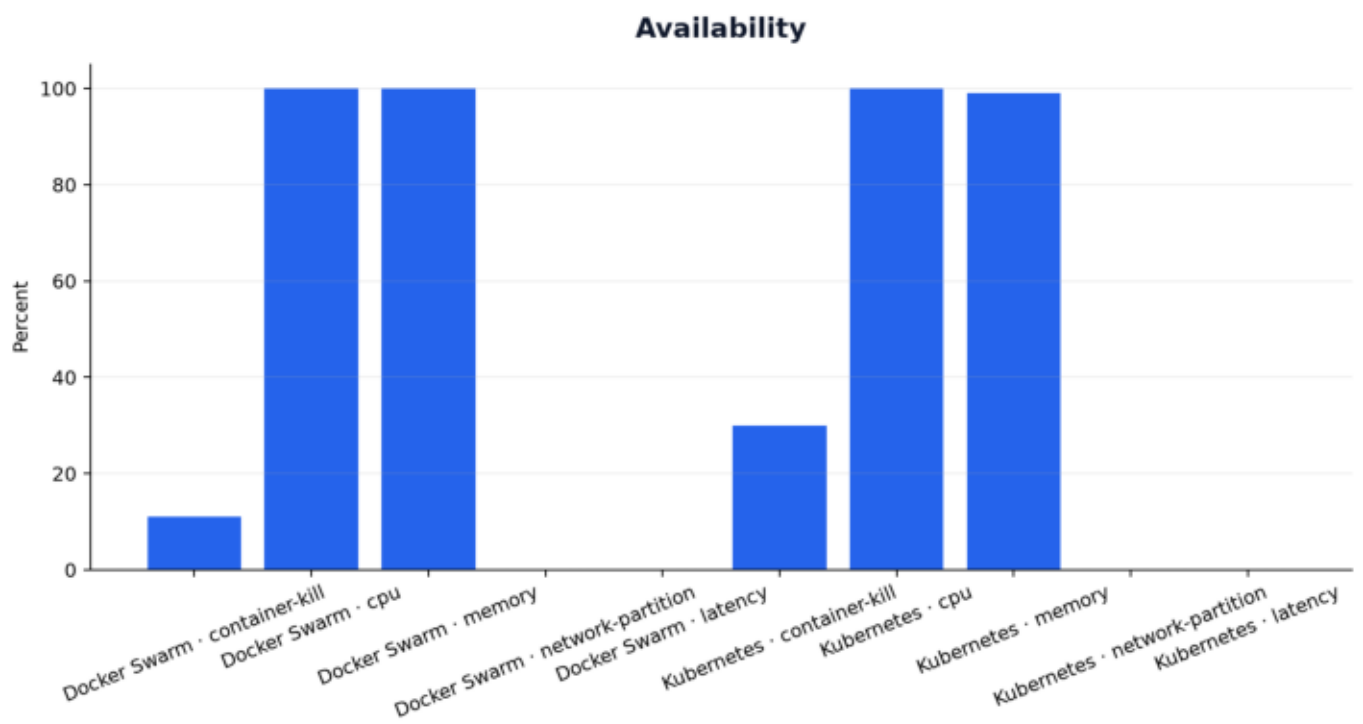
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

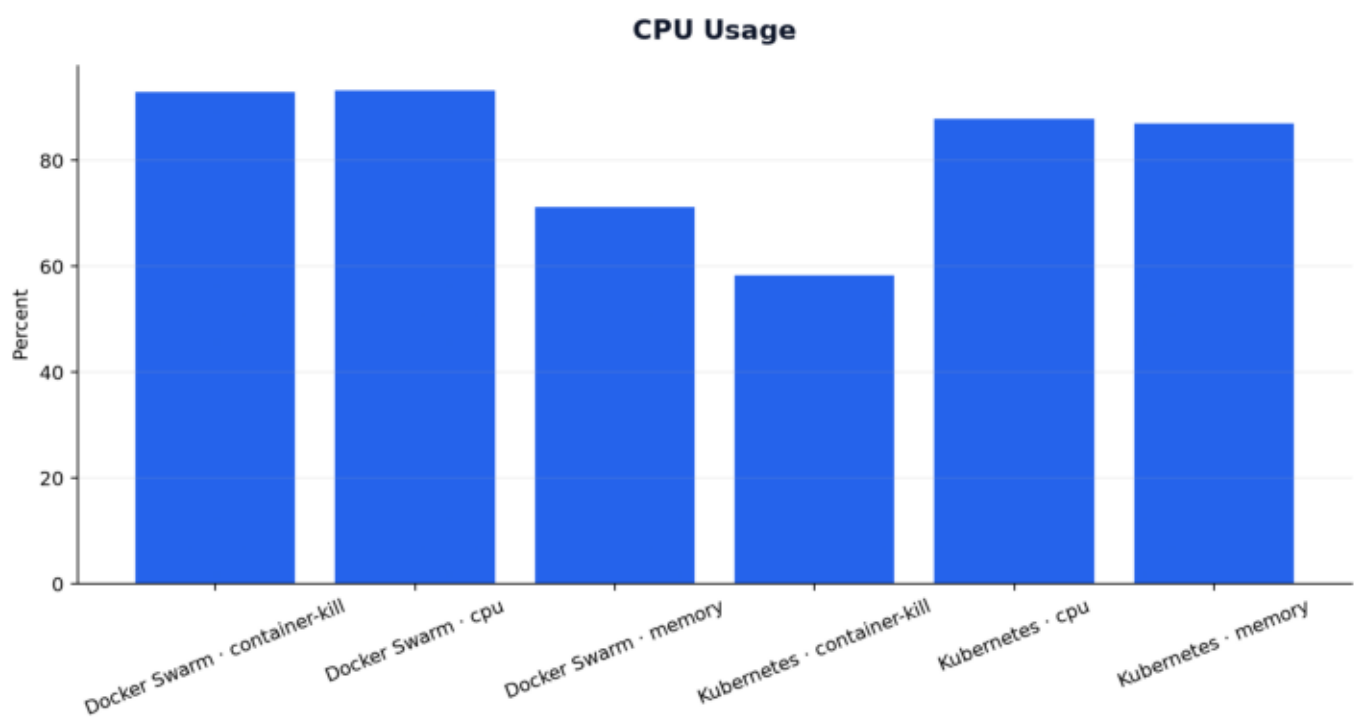
Availability

Current-session measured data



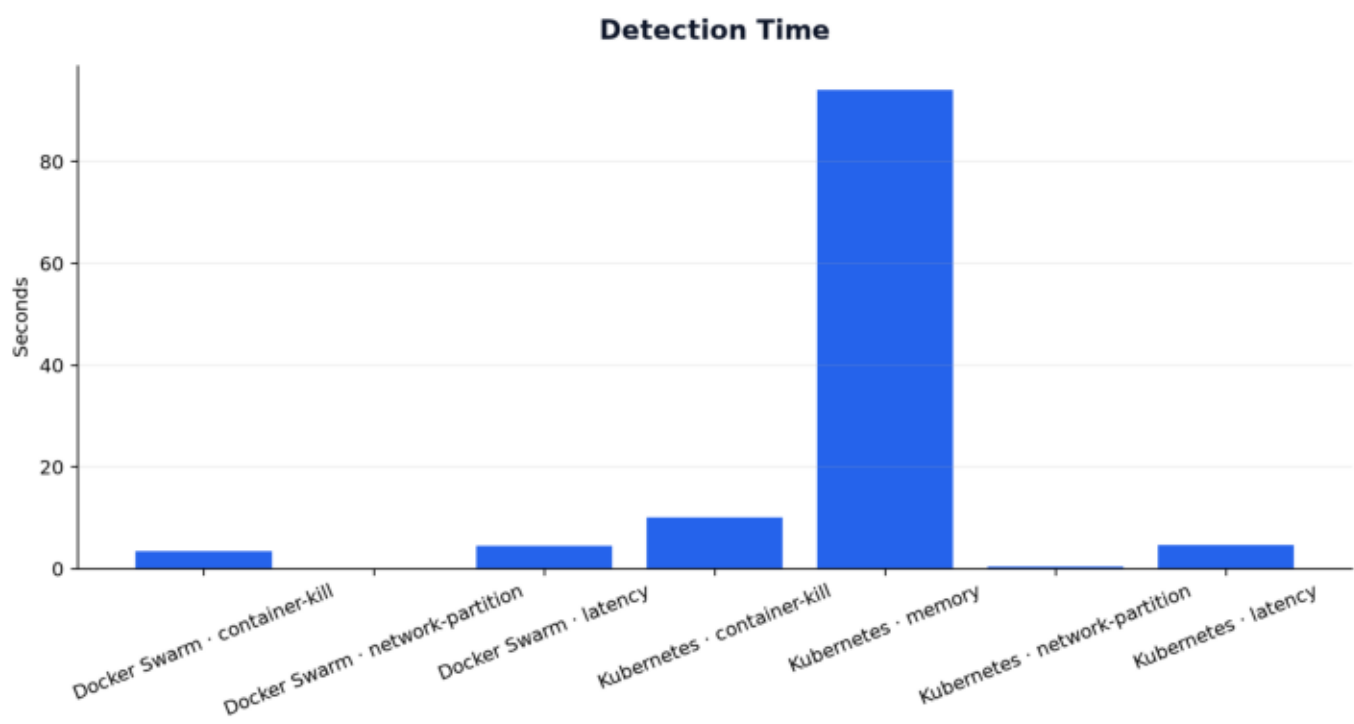
Cpu Usage

Current-session measured data



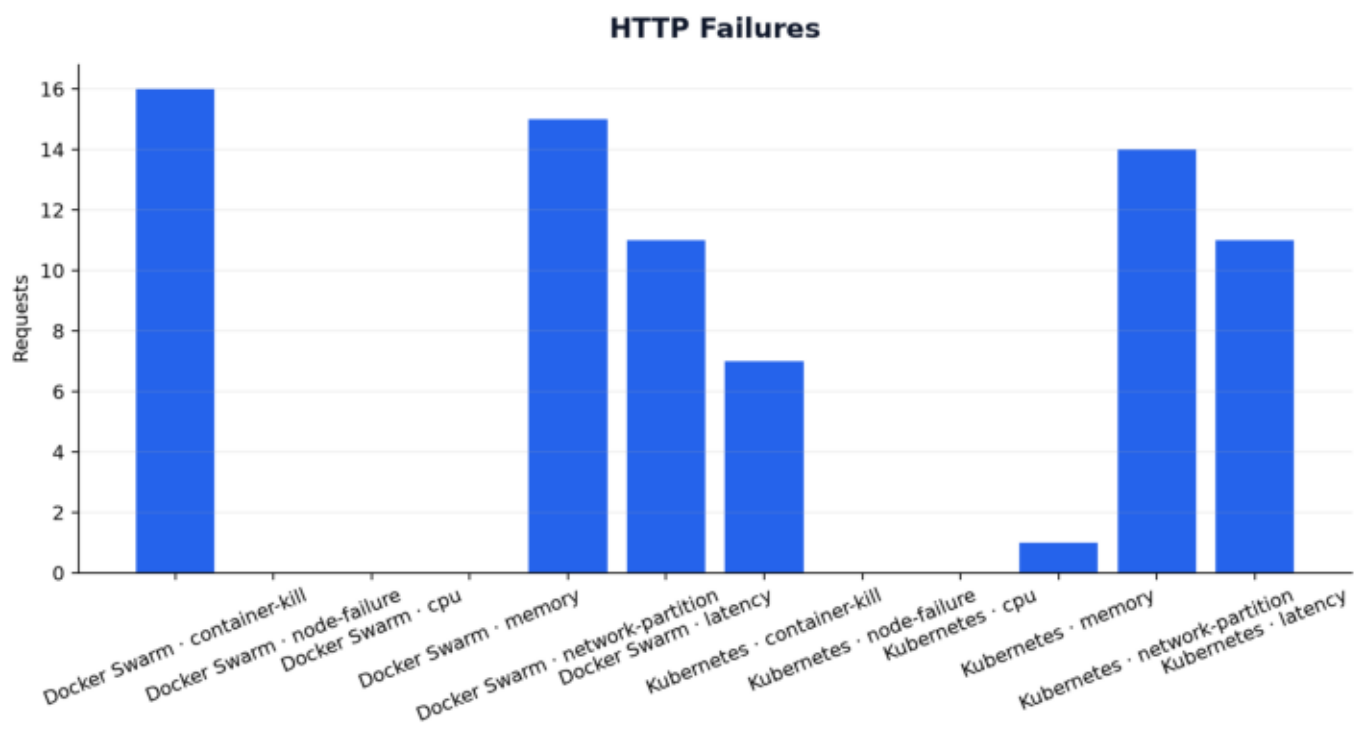
Detection Time

Current-session measured data



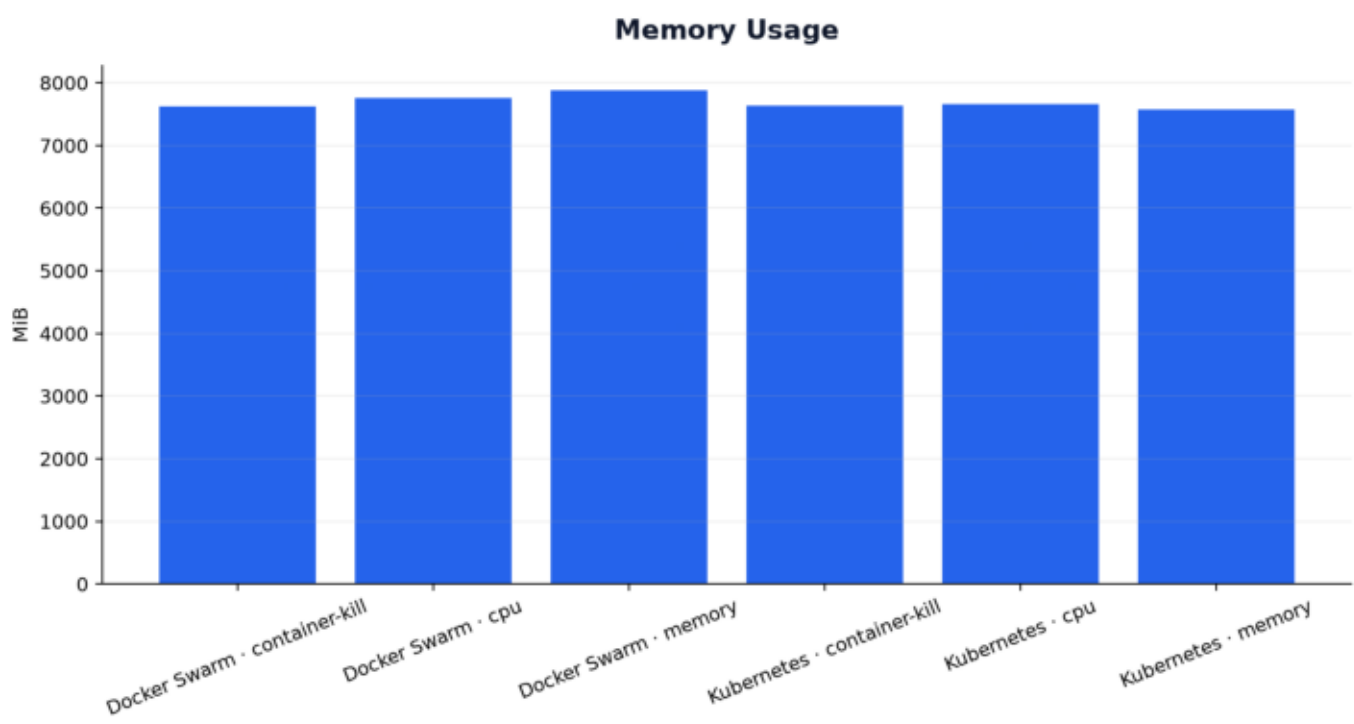
Http Failures

Current-session measured data



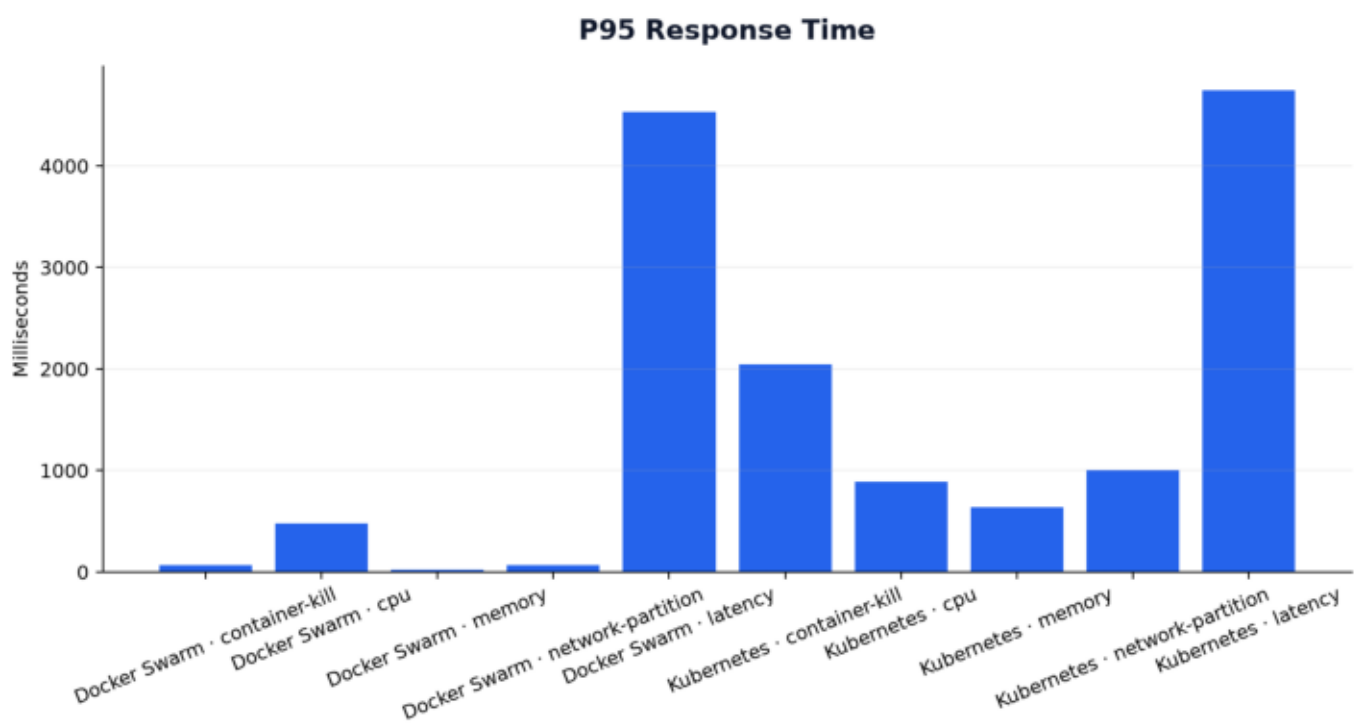
Memory Usage

Current-session measured data



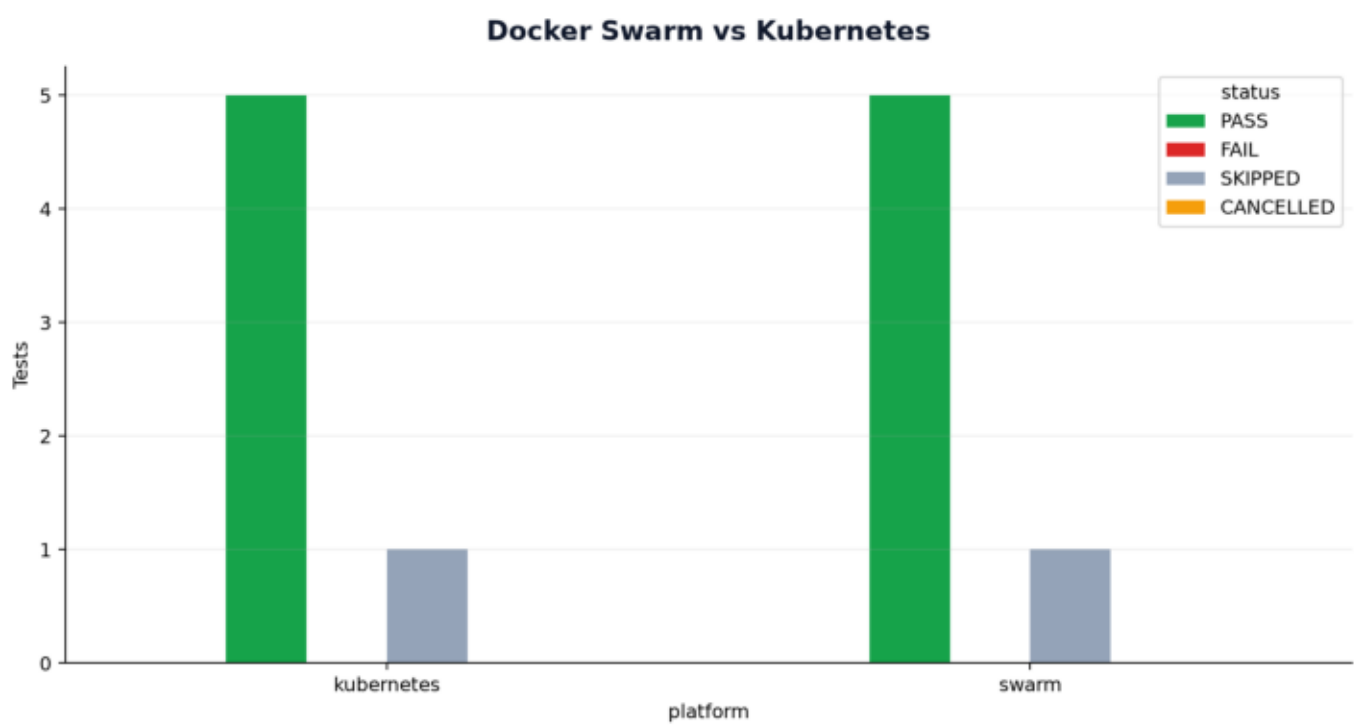
P95 Response Time

Current-session measured data



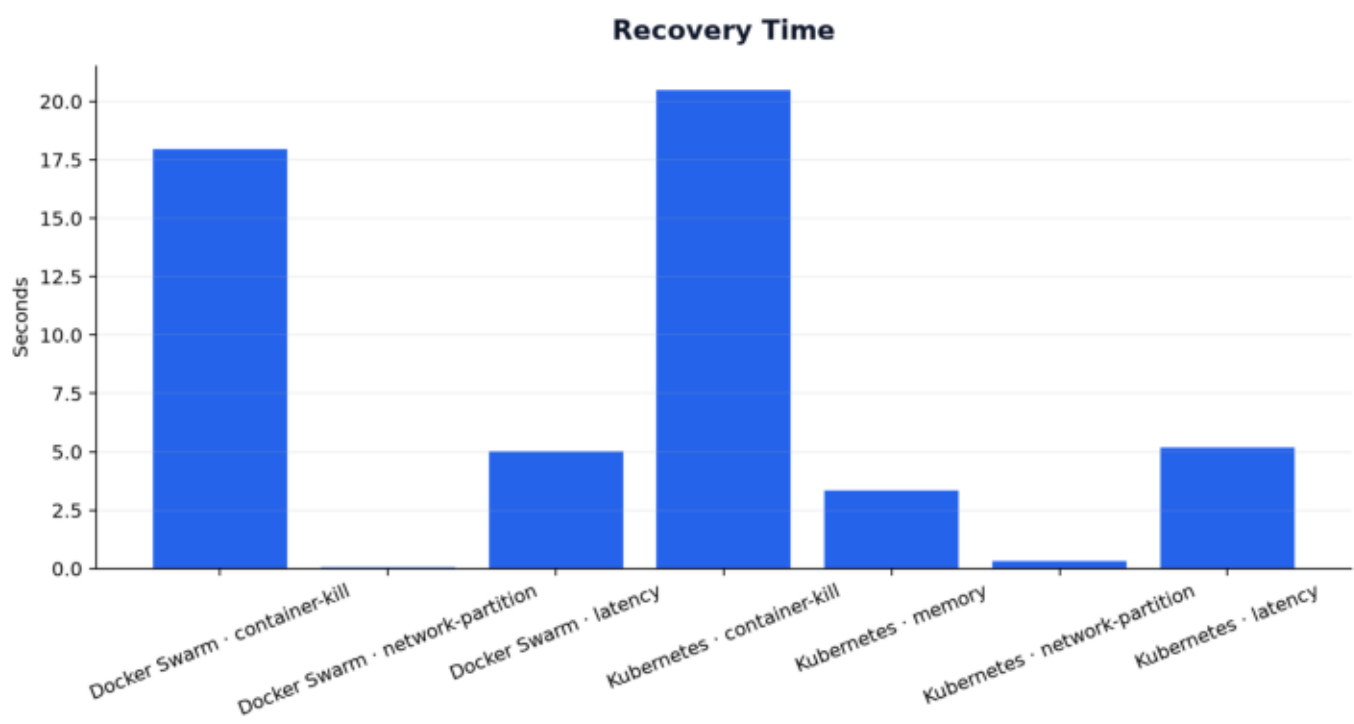
Platform Comparison

Current-session measured data



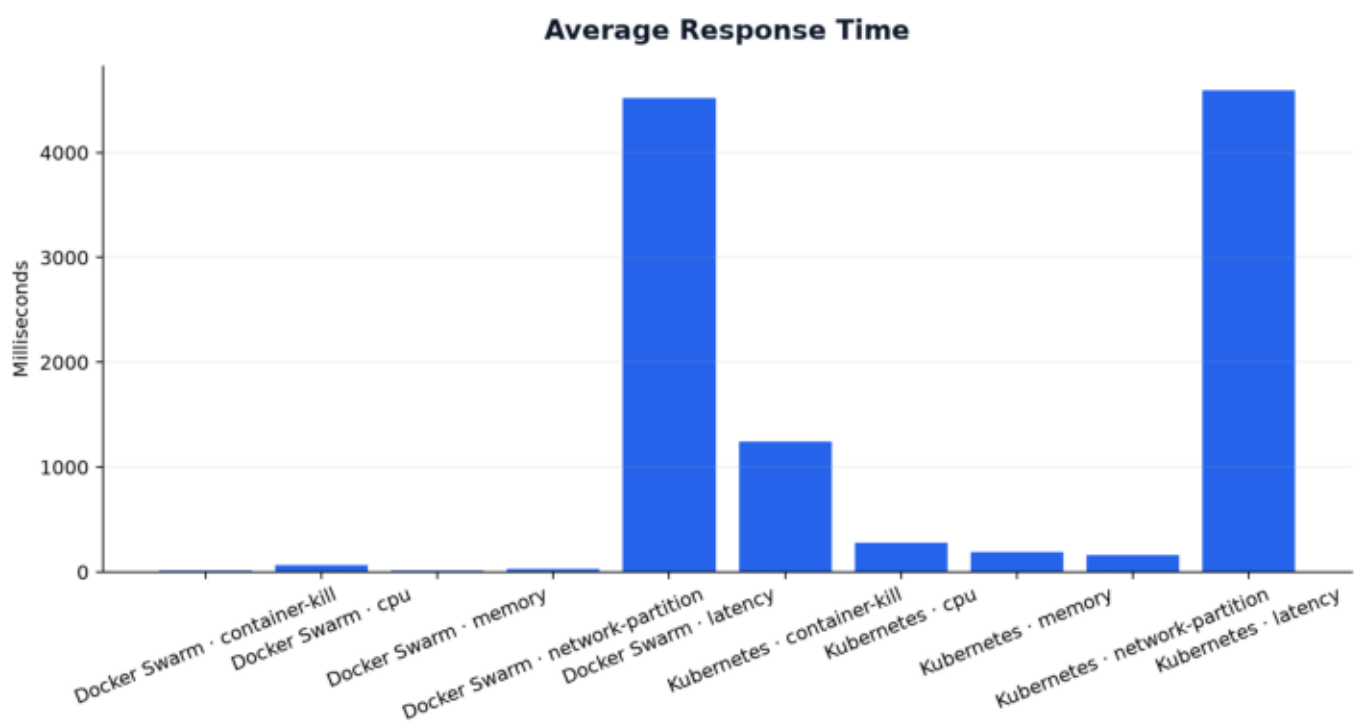
Recovery Time

Current-session measured data



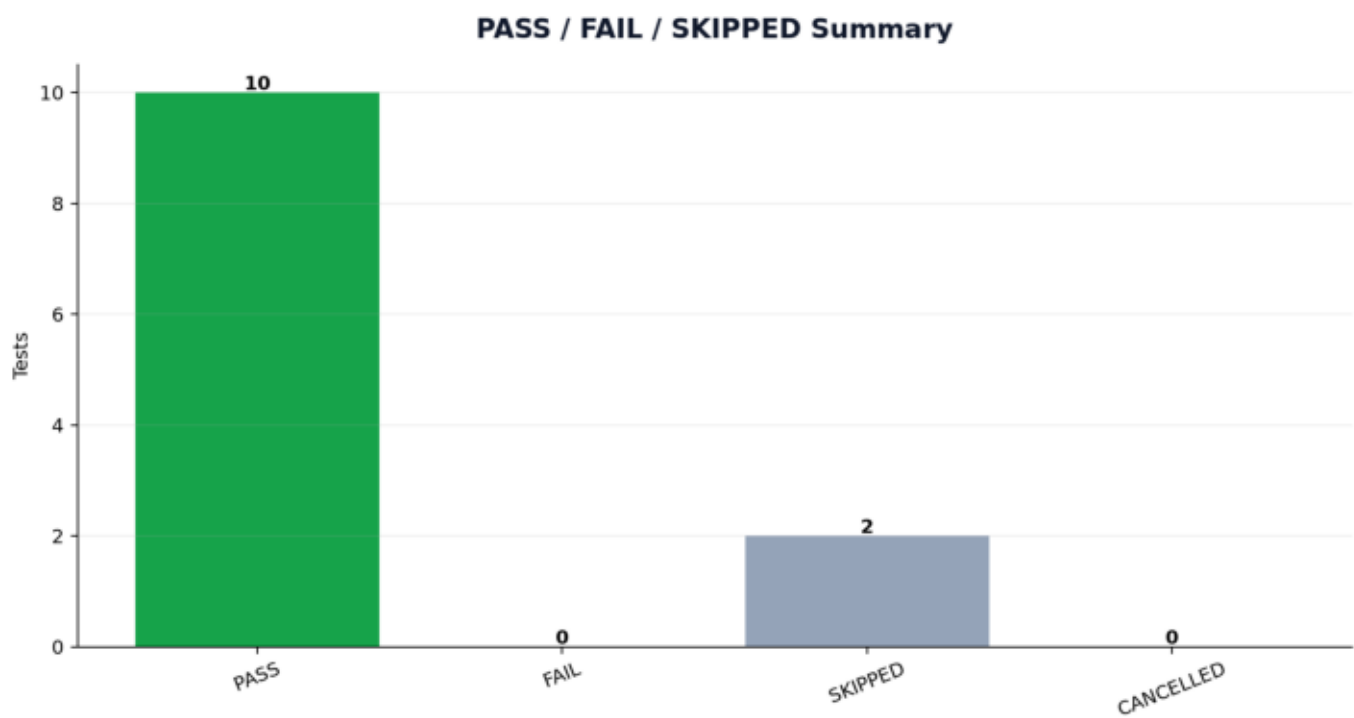
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
—	—	—	No failures recorded.

Skipped Tests

Platform	Scenario	Exact reason
swarm	node-failure	Single-node Docker Swarm cannot demonstrate real node rescheduling.
kubernetes	node-failure	Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

Swarm vs Kubernetes Comparison

Platform	PASS	SKIPPED
kubernetes	5	1
swarm	5	1

Final Conclusion

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During this campaign, 10 of 12 scenarios passed, 0 failed and 2 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.